Basics of Hypothesis Testing

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Outline

Test of Hypothesis - Overview

- A statistical hypothesis is a claim about a population parameter
- We test this claim using sample data
- The goal is to make a decision under uncertainty
- Decisions are made using probability and sampling distributions

Statistical Hypotheses

- **Null hypothesis (H_0):** baseline assumption
- **Alternative hypothesis (H_A):** competing claim
- They are mutually exclusive and exhaustive
- Example:

$$H_0 : \mu = 100 \quad \text{vs} \quad H_A : \mu \neq 100$$

- Based on the available data, we either reject the null hypothesis in favor of the alternative or "fail to reject the null hypothesis"
- Therefore, we do not "accept" the null hypothesis

Types of Alternative Hypotheses

- Two-tailed:

$$H_A : \mu \neq \mu_0$$

- Left-tailed:

$$H_A : \mu < \mu_0$$

- Right-tailed:

$$H_A : \mu > \mu_0$$

Test Statistic

- The test statistic function of sample data used to test H_0
- Example (Z-test):

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

- Decision is based on the sampling distribution under H_0

Type I Error

- Rejecting H_0 when it is true
- Probability of type I error (also known as *level of significance*):

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$$

- Also called false positive

Type II Error

- Failing to reject H_0 when it is false
- Probability of type II error:

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ false})$$

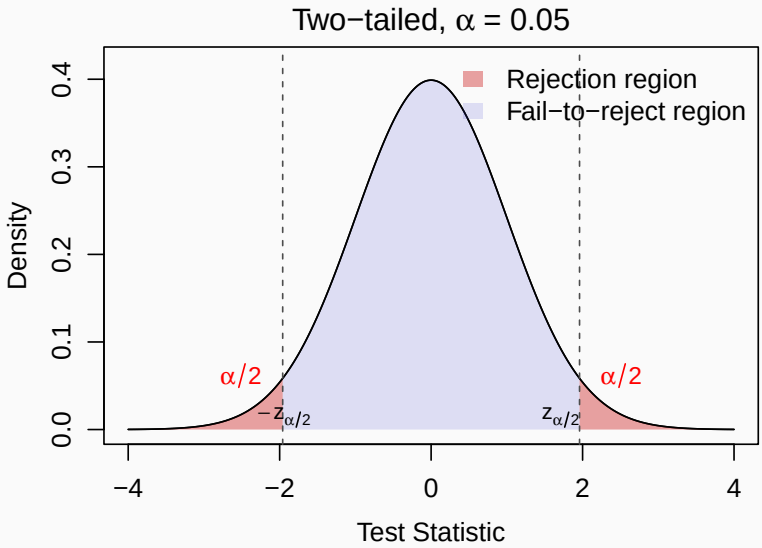
- Also called false negative

Level of Significance

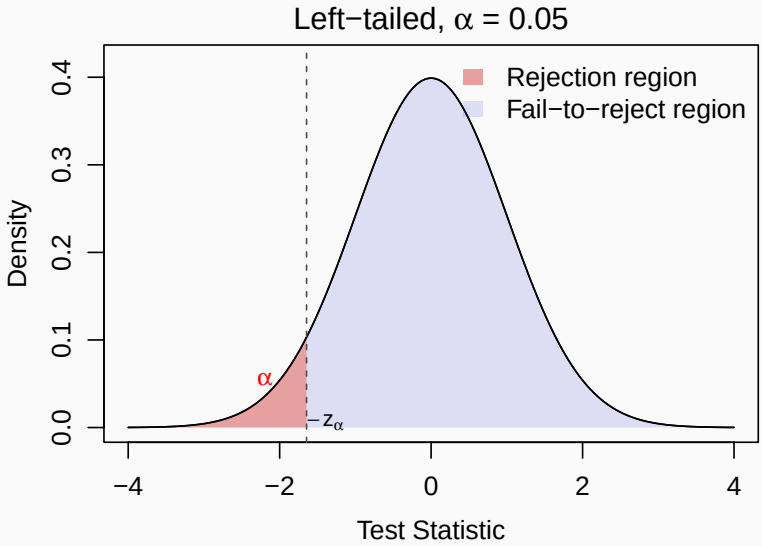
- $\alpha = P(\text{Type I error})$
- Common choices: $0.10,\ 0.05,\ 0.01$
- It is essentially our "tolerance" for type I error
- Smaller choice of α means stricter evidence required
- Confidence level:

$$1 - \alpha$$

Rejection Region and α in Two-tailed Tests



Rejection Region and α in Left Tests



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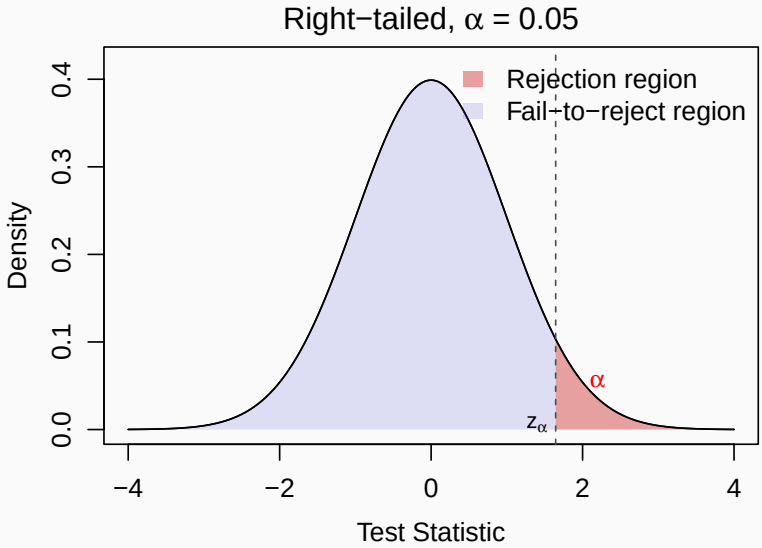


Power of a Test

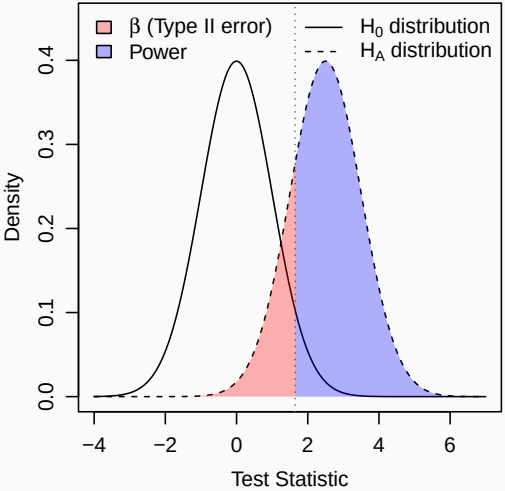
- **Power** is the probability of correctly rejecting H_0 when it is actually false

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ false})$$

- A test with low power may fail to reject a false H_0 which is a *false negative* error
- Power depends on:
  - **Sample size** n: larger $n \Rightarrow$ higher power
  - **Effect size** $|\mu - \mu_0|$: larger effect $\Rightarrow$ easier to detect
  - **Variability** σ: smaller $\sigma \Rightarrow$ higher power
  - **Significance level** α: larger $\alpha \Rightarrow$ higher power (but at the cost of higher Type I error probability)
- Conventionally, power ≥ 0.80 is considered adequate

Power: The Two Overlapping Distributions

- Under H_0: test statistic is centered at μ_0
- Under H_A: test statistic is centered at some true $\mu_A \neq \mu_0$
- The **rejection region** is fixed by α using the H_0 distribution
- **Power** is the area of the H_A distribution that falls *inside* the rejection region
- β is the area of the H_A distribution that falls *outside* the rejection region



Power: The α–β Trade-off

- For a fixed sample size, reducing α (stricter test) **moves the rejection region further into the tail**
- This means less of the H_A distribution falls in the rejection region $\Rightarrow$ power **decreases** and β **increases**
- There is an unavoidable trade-off:

α	Type I Error	Power $(1-\beta)$
Smaller	Lower	Lower
Larger	Higher	Higher

- The only way to reduce *both* errors simultaneously is to **increase the sample size**
- In practice, α is fixed first (usually 0.05), then n is chosen to achieve the desired power

Errors Summary Table

	H_0 True	H_0 False
Reject H_0	Type I (α)	Correct ($1-\beta$)
Fail to reject H_0	Correct ($1-\alpha$)	Type II (β)

How a Test is Performed

- Suppose we have a hypothesis $\mu_x = \mu_0$ regarding a random variable X
- We collect the data and compute the sample mean $\bar{x}$
- We either "reject" or "fail to reject" the H_0 based on how "far" $\bar{x}$ is from μ_0
- In order to be able to quantify the distance between $\bar{x}$ and μ_0 in a statistically meaningful way, we have to "convert" the sample mean to a test statistic
- Then based on the value of the test statistic and the sampling distribution of the test statistic, a decision regarding the null hypothesis is made

Sampling Distributions of Test Statistics

- Different test statistics follow different kinds of sampling distributions
- Hypothesis testing relies on those sampling distributions
- Assuming that H_0 is true, the test statistic follows a known sampling distribution
- Example: $Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$
- This allows us to quantify "how far" $\bar{X}$ is from μ_0, in other words, "how unusual" our sample is if the null hypothesis is to be true
- If the test statistic indicates that the data in the sample is too unusual for the null hypothesis to be true, then the null hypothesis is rejected

Rejection Region and Critical Value

- Define a region in the distribution of the test statistic where H_0 is rejected
- Based on level of significance α
- For two-tailed test:

$$P(\text{Reject } H_0 \mid H_0 \text{ true}) = \alpha$$

General Testing Procedure

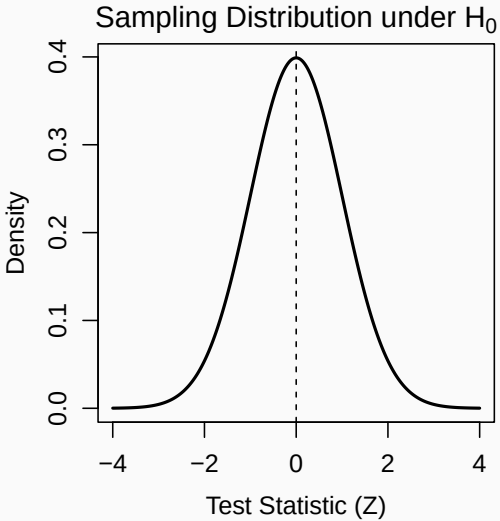
- State H_0 and H_A
- Choose α
- Select test statistic
- Determine the rejection region
- Compute the value of the test statistic
- Make decision: if the test statistic falls in the rejection region then H_0 is rejected

Sampling Distribution under the Null Hypothesis

Suppose the plot shows the sampling distribution of some test statistic Z.

Different test statistics have different sampling distribution.

After obtaining the distribution of the test statistic, a region is determined where the calculated value of the test statistic is unlikely to appear.

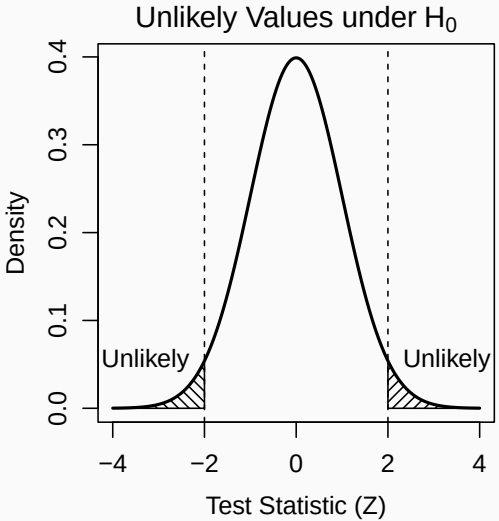


Unlikely Region of the Test Statistic under the Null Hypothesis

The highlighted region in the plot shows a region of in sampling distribution of the test statistic in which the calculated value of the test staistic is unlikely to fall.

The question remains as to how to determine this region.

This region is determined using the level of significance.



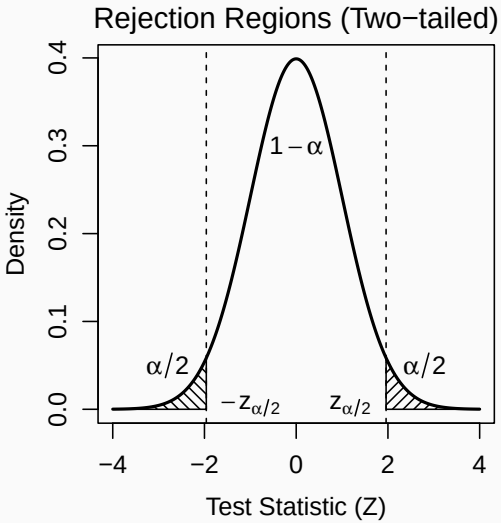
Rejection Region in Two-tailed Tests

Suppose a value of the "level of significance, α" is chosen.

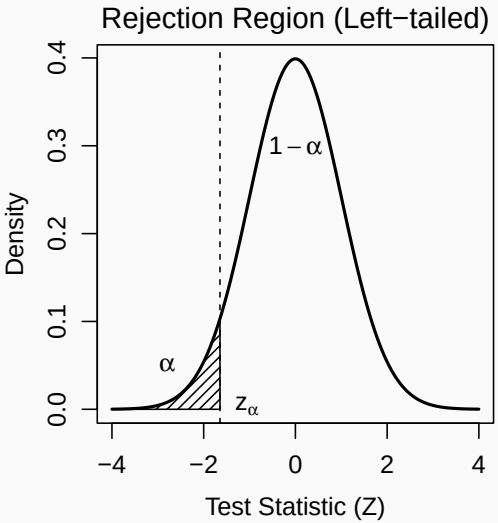
Then the rejection region is chosen such that the area in the rejection region equals to α.

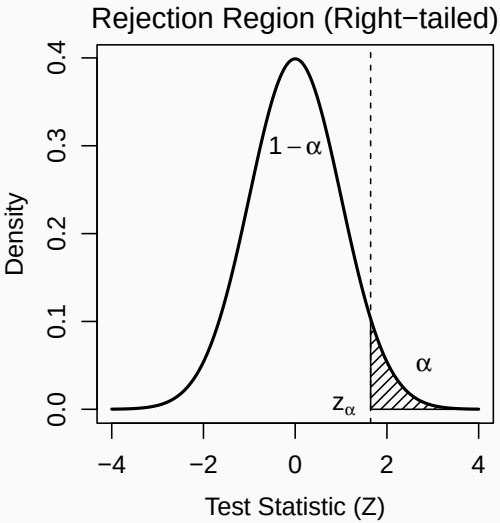
Since this is a two-tailed test, the rejection region is on both sides of the distribution and each region has area equal to $\alpha/2$.

If the calculated value of the test statistic falls in this region, then the null hypothesis rejected.



Rejection Region in One-tailed Tests





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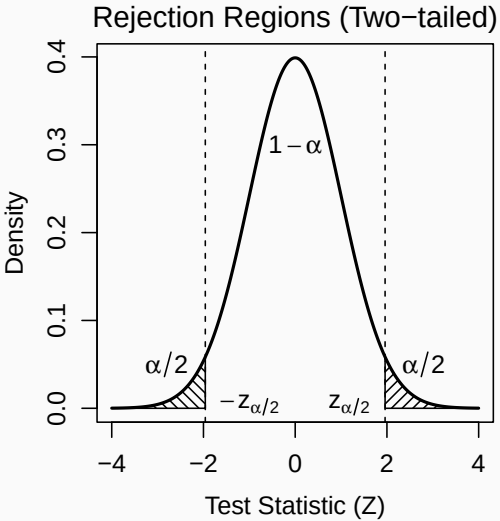
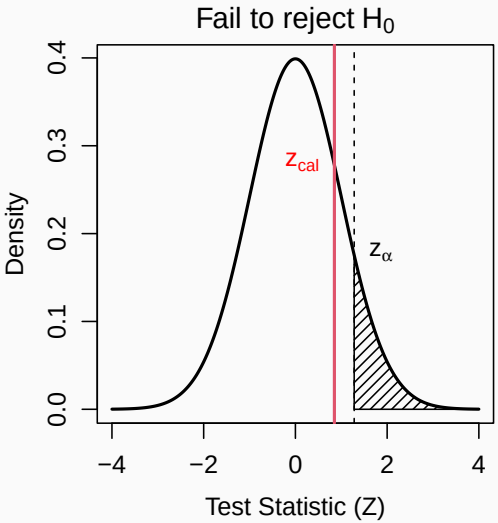
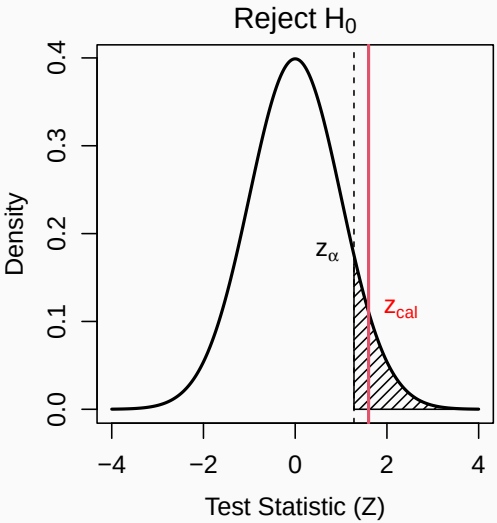


Decision Based on Calculated Value of the Test Statistic





Example Setup

- Claim: teens sleep more than 10 hours
- Sample mean: $\bar{x} = 11.5$
- $\sigma = 5$, $n = 20$
- Hypotheses:

$$H_0 : \mu = 10 \quad H_A : \mu > 10$$

Example Test Statistic

- Test statistic:

$$Z = \frac{11.5 - 10}{5/\sqrt{20}} = 1.34$$

Example Decision

- At $\alpha = 0.05$, the critical value from the sampling distribution of the test statistic, $z_{0.05}$ is equal to 1.64
  - Since $1.34 < 1.64$, we fail to reject H_0
- At $\alpha = 0.10$, the critical value from the sampling distribution of the test statistic, $z_{0.10}$ is equal to 1.28
  - Since $1.34 > 1.28$, we reject H_0

P-value

- The **p-value** is the probability of observing a test statistic as extreme as, or more extreme than, the one calculated from the sample — *assuming H_0 is true*
- It is **not** the probability that H_0 is true
- It quantifies how "surprising" the data are under H_0
- Formally, for a right-tailed test:

$$p = P(Z \geqslant Z_{\text{cal}} \mid H_0 \text{ true})$$

- Decision rule:
  - $p \leqslant \alpha$: reject H_0
  - $p > \alpha$: fail to reject H_0

P-value Interpretation

- A **small** p-value means the observed data would be very unlikely if H_0 were true, which is evidence against H_0
- A **large** p-value means the data are consistent with H_0, but this does *not* prove H_0 is true. Therefore, we do not "accept" the H_0, rather we "fail to reject it"
- Common thresholds (informal):
  - $p < 0.10$: some evidence against H_0
  - $p < 0.05$: strong evidence against H_0
  - $p < 0.01$: very strong evidence against H_0
- **Common misconceptions:**
  - p-value is *not* the probability the result occurred by chance
  - p-value is *not* a measure of effect size or practical significance
  - A non-significant p-value does not "accept" H_0

P-value: Tails and Computation

- The tail(s) used depend on the alternative hypothesis H_A:
  - Right-tailed ($H_A : \mu > \mu_0$): $p = P(Z \geq Z_{\text{cal}})$
  - Left-tailed ($H_A : \mu < \mu_0$): $p = P(Z \leq Z_{\text{cal}})$
  - Two-tailed ($H_A : \mu \neq \mu_0$): $p = 2\,P(Z \geq |Z_{\text{cal}}|)$
- **Example** (from earlier): $Z_{\text{cal}} = 1.34$, right-tailed test

$$p = P(Z \geq 1.34) \approx 0.090$$

- Since $p = 0.090 > 0.05 = \alpha$, we fail to reject H_0 at 5% of significance
- Since $p = 0.090 < 0.10 = \alpha$, we reject H_0 at 10% level of significance

Confidence Interval: Construction and Meaning

- A $100(1-\alpha)\%$ CI is an interval constructed from sample data such that, in repeated sampling, it contains the true parameter $100(1-\alpha)\%$ of the time
- **Correct interpretation:** if we drew many samples and built a CI from each, $(1-\alpha)$ of those intervals would contain μ
- **Incorrect interpretation:** "there is a $(1-\alpha)$ probability that μ lies in *this* interval" — μ is fixed; probability here is a property of the *procedure*
- Width of CI:

$$\text{Width} = 2\, z_{\alpha/2}\, \frac{\sigma}{\sqrt{n}}$$

- Wider CI: less precision; Narrower CI: more precision
- Trade-off: higher confidence level $(1-\alpha)\uparrow \Rightarrow$ wider interval

Connection with Confidence Intervals

- A $100(1-\alpha)\%$ confidence interval for μ is:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- A CI gives a **range of plausible values** for μ, whereas a hypothesis test gives a binary decision
- The duality: for a two-tailed test at level α,

$$\text{Reject } H_0 : \mu = \mu_0 \iff \mu_0 \notin \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- **Advantage of CI over hypothesis test:** a CI also communicates the magnitude and precision of the estimate, not just whether to reject H_0
- Narrower CI $\Rightarrow$ more precise estimate (achieved by larger n or smaller σ)

Key Takeaways

- Hypothesis testing is decision-making under uncertainty
- α controls Type I error
- β and power describe sensitivity
- P-value provides flexible decision rule
- Sampling distributions are the foundation

Thank you.